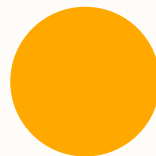




Programming Languages

Reese Hatfield



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Programming Languages

- You know have the ability to write anything in Javascript*
- Can go even higher level
 - Define algorithms
 - Pseudocode
- But let's look at the leap in logic





Programming Languages

- Huge gap between hardware and software
- We went from constructing gates
 - To building web pages
- We skipped about 40+ years of history
- Programming hasn't always been like this





Programming Languages

- Javascript
 - Programming Language
 - Built for websites
- Call JS a *high* level programming languages
- It's very difficult to figure out what our code does at a hardware level





Programming Languages

- Don't forget
 - Every line of code we write
 - Corresponds to specific input
 - In some form of hardware
 - Just a bunch of gates
 - Just a bunch of 1's and 0's





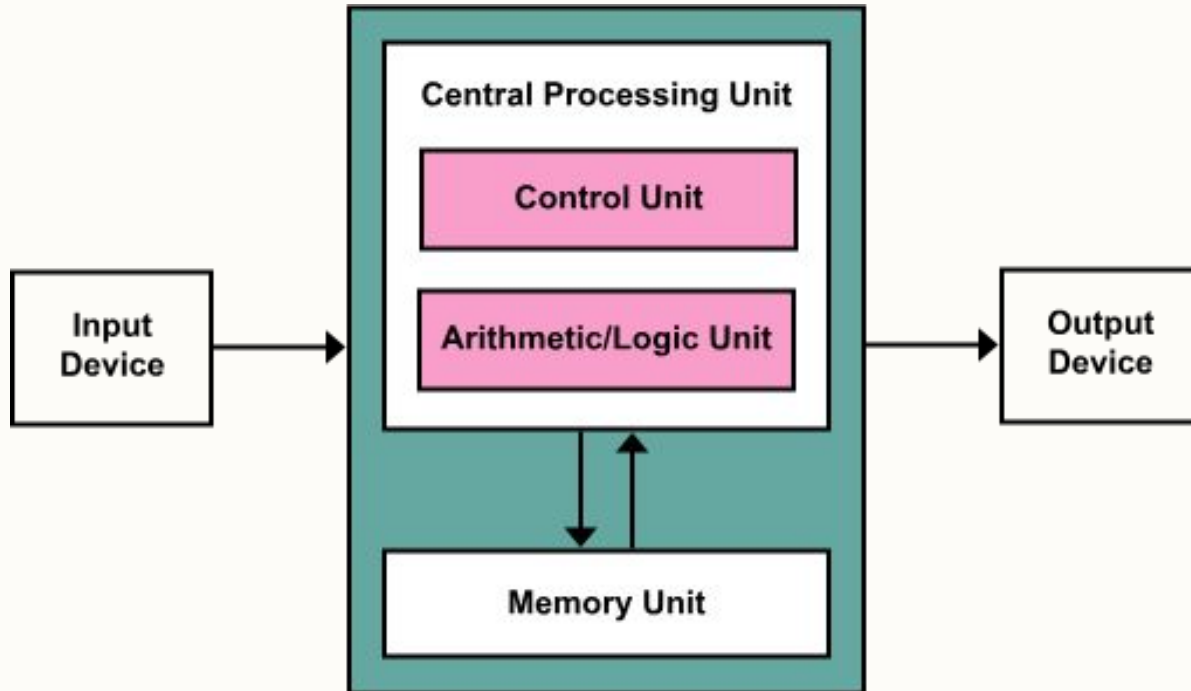
Programming Languages

- So how did this gap get bridged in the first place
- Let's revisit the idea of a CPU
 - Instruction Processor
- To understand this gap, we first need to understand CPU architecture



Programming Languages

Von Neumann Architecture





Programming Languages

- Receive arbitrary input (Instructions)
- Tell me how to manipulate memory
 - Fast registers
- Literally
 - Move this data into Register 1
 - Add Reg. 1 with Reg 2, store result in Reg 3.





Programming Languages

- At a *very* low level
- And always at an earlier point in time
- Pseudo code looked like this:
 - Move this data into Register 1
 - Add Reg. 1 with Reg 2, store result in Reg 3.
- CPUs still work like this
 - At their core





Programming Languages

- But how can you program like this?
- This idea comes from **assembly programming languages**
- Specific to any given CPU / manufacturer*
- Can be translated DIRECTLY into binary



Programming Languages

- So let's see what this looks like
- x86 ISA
- Left=Hex (binary)
- Right= code

```
08048918    pushl    %ebp
08048919    movl     %esp,%ebp
0804891b    subl     $0x4,%esp
0804891e    movl     $0x0,0xffffffff(%ebp)
08048925    cmpl     $0x63,0xffffffff(%ebp)
08048929    jle      08048930
0804892b    jmp      08048948
0804892d    nop
0804892e    nop
0804892f    nop
08048930    movl     0xffffffff(%ebp),%eax
08048933    pushl    %eax
08048934    pushl    $0x8049418
08048939    call     080487c0 <printf>
0804893e    addl     $0x8,%esp
08048941    incl     0xffffffff(%ebp)
08048944    jmp      08048925
08048946    nop
08048947    nop
08048948    xorl     %eax,%eax
0804894a    jmp      0804894c
0804894c    leave
0804894d    ret
```



Programming Languages

- This is *really* difficult to program in
- I have no idea what that example is doing
- I have done assembly before
 - You can too!
- Your CPU knows exactly how to process that tho!





Programming Languages

- Building software like this does not scale
- How do you make a website???
- Possible, but not practical
- Few notable exceptions
 - Roller Coaster Tycoon





Programming Languages





Programming Languages

- So what happened?
 - How did we go from assembly to javascript?
- What if we did this:
 - Write an assembly programming
 - That read different instructions
 - But ultimately converted those high level instructions back into ASM





Programming Languages

High Level
Program



Assembly
(Pseudocode)

int i = 10;



Clear REG 1;
Associate label "i" with REG 1;
Load 0b1010 into REG 1





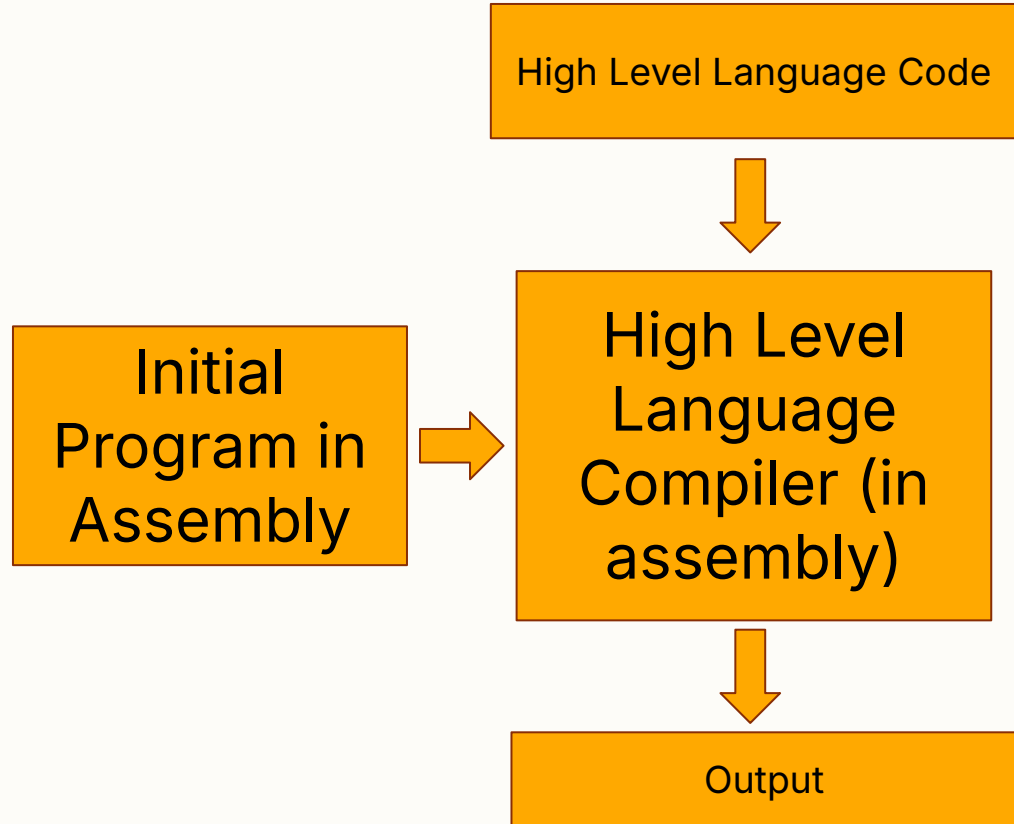
Programming Languages

- First big “high level” languages
 - **C** (important one)
 - **FORTRAN**
 - **COBOL**
- **C** is very simple (lets see examples.)
- Has a simple Compiler
- Easy to turn into assembly
 - Relatively



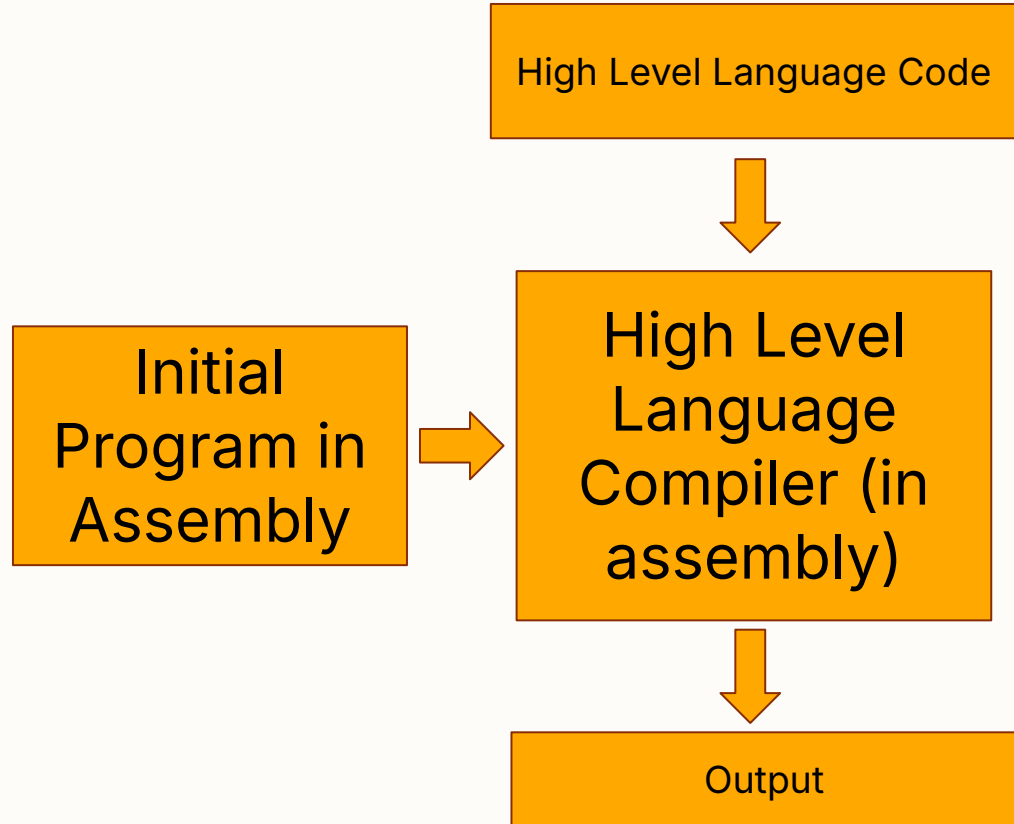
Programming Languages

- Write a **"compiler"**:
- *A program that turns one programming language into another*



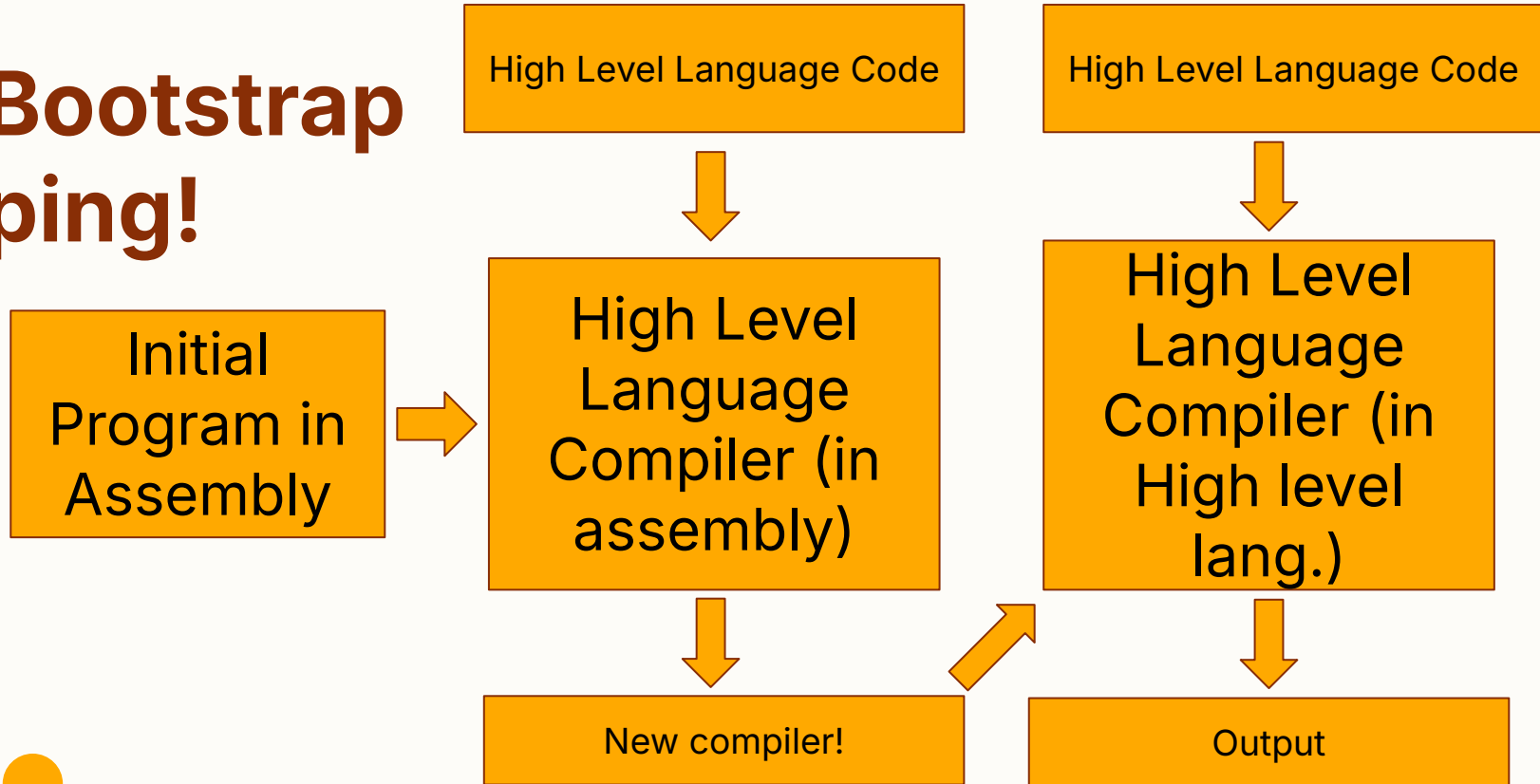
Programming Languages

- If our new language is turing complete, we can write anything in it
- Let's write ANOTHER compiler



Programming Languages

Bootstrapping!





Programming Languages

- "What language is the language written in?"
- Answer is usually itself!
- Java $\Rightarrow$ Written in Java*
- C $\Rightarrow$ Written in C*
- etc.





Programming Languages

- This process continues
- Onward and onwards
- Until you end up with languages like Javascript and python
- Takes a long time
- New languages at different levels are being written everyday
 - Field of study





Programming Languages

- Language Paradigms
- All the languages we have seen so far have been relatively simple to read
- They are **procedural** languages





Programming Languages

- **Procedural Languages**
- Very common
- "Perform a concrete, series of instructions upon the data"
- Lends itself well to assembly code





Programming Languages

- **Object Oriented Languages**
- Model your program as objects in your code
- Mutate the objects (data) in a particular way
- Java and Ruby are two languages that come to mind





Programming Languages

- **Functional Languages**
- Define a series of concrete functions that act upon data, with consistent I/O
- Languages that come to mind are Lisp, Haskell, Racket :(





Programming Languages

- Almost every language is multi-paradigm
- They just lean into certain ones harder
- It's very difficult to write code in only one paradigm
- They all have benefits in different scenarios





Programming Languages

- Computer Science requires programming
- Fundamental to the discipline
- Almost inseparable
 - See "Algorithms"

